Ortak Hesap

Design justification · Milestone 1 · shared-expense app for students

Team 5 · MTM4681 Mobile Programming
Hakan İnce & İlhan Ünal
Flutter · Material 3

1 Product, users and design principles

Ortak Hesap helps students who share a flat, a dorm room or a holiday keep track of who paid for what and who owes whom. Our users are 18–26 years old, use the phone one-handed (in a shop, a café, on the bus), often in the evening, and are not accountants: they want to log a purchase in a few seconds and see one clear answer, "how much do I owe or get back?". Money between friends is also socially sensitive, so the interface must feel calm and fair rather than alarming. This led to three principles:

Speed to add
Amount first, sensible defaults (paid by me, split equally), one primary action per screen.

Clarity of money
One headline balance per group, who-owes-whom in plain words, never colour alone.

Calm, fair tone
Friendly teal and warm terracotta instead of banking blue and alarm red.

2 Colour palette

We built a Material 3 scheme from the seed colour `#0B6B63` (deep teal) and tuned two roles by hand. Teal signals trust and money but stays distinct from the blue of most banking apps, and reads as "growth" for money you are *owed*. Debts use a warm terracotta (tertiary role) instead of red: red is reserved for errors and destructive actions, so a normal debt does not feel like a warning.

Swatch	M3 role	Hex	Used for	Pair checked	Contrast
	primary	`#0B6B63`	FAB, buttons, "owed to you" amounts	on surface #F5FAF8	**6.0 : 1**
	primaryContainer	`#B8F0E7`	Selected tab, summary cards	text #00201C	**13.6 : 1**
	tertiary	`#9A4521`	"you owe" amounts, pending settlements	on surface #F5FAF8	**6.1 : 1**
	tertiaryContainer	`#FFDBCC`	Debt chips, info banners	text #3A0C00	**13.2 : 1**
	surface	`#F5FAF8`	Screen background	text #171D1C	**16.2 : 1**
	outline	`#6F7977`	Input borders, dividers	UI component (3 : 1)	**4.3 : 1**
	error	`#BA1A1A`	Validation errors, delete	on surface #F5FAF8	**6.1 : 1**
	dark theme	`#0E1514` / `#82D5CA` / `#FFB596`	Surface / primary / tertiary in dark mode	primary & tertiary on surface	**10.8 : 1**

Colour is never the only signal. Teal and terracotta have almost the same luminance, so a colour-blind user (or a screenshot in greyscale) could confuse them. Every amount therefore also carries a sign (+ / −), an arrow icon and a text label ("owes you" / "you owe").

3 Typography

Plus Jakarta Sans (SemiBold/Bold) is used for display, headline and title styles. Its geometric, slightly rounded letterforms feel friendly and modern, which suits a social-money app and gives the group names and balances character. **Inter** is used for body, labels and every amount: it is designed for small screens, stays legible at 12 sp and offers *tabular figures*, so digits line up in columns of amounts. Both families fully support Turkish characters (ç ğ ı İ ö ş ü) and the lira sign, which matters for our users.

We follow the Material 3 type scale with sizes in **sp**, so text grows with the user's system font setting (the layouts are designed to survive 200 %). The smallest size is 12 sp, used only for captions.



Scale used: Display Small 36/44 sp (group balance) · Title Large 22/28 (screen titles) · Title/Body Large 16/24 (list rows, input text) · Body Medium/Label Large 14/20 (secondary text, buttons) · Label Small 12/16 (captions, tab labels).

4 Layout and navigation

- **Thumb-first, compact layout.** We design for a 360×800 dp phone first. A Material 3 *navigation bar* with three destinations (Groups, Activity, Profile) sits at the bottom, and the extended *FAB "Add expense"* is placed bottom-right, inside the natural thumb zone. The most frequent action is therefore one tap away from any group screen.
- **Progressive disclosure.** The Add-expense form opens with only amount, title and payer; the split defaults to "equally" and the custom-split sheet appears only on request.
- **Consistent grid and shapes.** 8 dp grid, 16 dp margins, 48×48 dp minimum touch targets, M3 corner radii (12 dp cards and fields, 28 dp FAB and sheets), tonal surfaces instead of heavy shadows.
- **Plain-language balances.** The Balances screen lists the minimal set of payments ("Ayşe pays Mert ₺120,00"), each with one "Settle" button and a confirmation dialog, because paying back is the goal of the app.
- **Helpful states and local formatting.** Empty states give one next step, errors appear inline with an icon and text, finished groups show "All settled". Amounts follow Turkish conventions (₺1.250,50) and texts stay short for TR/EN.

5 Accessibility

Area	Decision
Contrast	All text/background pairs meet WCAG 2.1 AA (≥ 4.5 : 1; most ≥ 6 : 1) and UI components such as borders meet 3 : 1. The dark theme is checked separately (10.8–14.3 : 1).
Not colour alone	Debt direction is expressed by sign, arrow icon and wording as well as colour; errors use an icon and message text.
Touch targets	Every interactive element is at least 48×48 dp with 8 dp spacing; list rows are fully tappable.
Scalable text	Sizes in sp, layouts use flexible containers and wrap instead of truncating; at 200 % scale, amount rows stack vertically (planned wireframe variant).
Screen readers	Flutter `Semantics` labels read full sentences ("Ayşe owes you 120 Turkish lira") rather than icon names; meaningful reading order top to bottom; decorative icons are excluded.
Motion	Transitions are short (200–300 ms) and respect the system "reduce motion" setting by falling back to simple fades.
Theme	Light and dark themes are both supported and follow the system setting, which helps in the evening and one-handed outdoor use.

6 How the design supports the intended users

Students open the app in short bursts, so the design favours recognition over reading: one large balance, Material 3 components they already know from Android, and a gentle colour scheme that does not make a debt feel like an accusation. In the prototype, motion is used sparingly and with purpose: the FAB expands into the Add-expense sheet (container transform) so users see where the form comes from, a new expense slides into the list, and the balance counts to its new value after a settlement, confirming that the action worked.